المعهد الهنـدي للتكنولوجيـــا دلهي
أبـــوظبي

INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

Evolution

Lecture 3

References:
CAMPBELL B I O L O G Y
(9th Edition)
Section; Unit 4, Chapter 22

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IIT Delhi

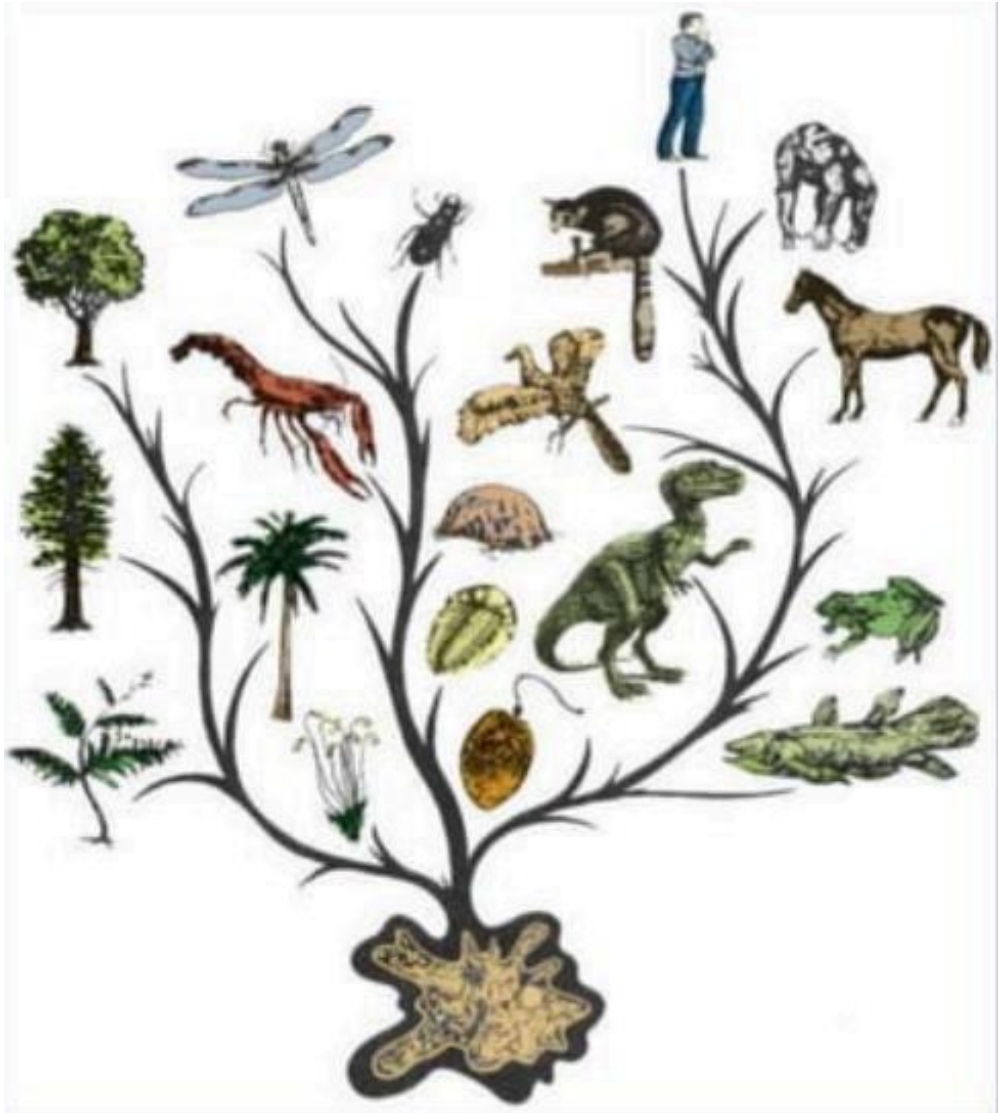
"Nothing in biology makes sense except in the light of evolution."

Theodosius Dobzhansky



thelogicofscience.com

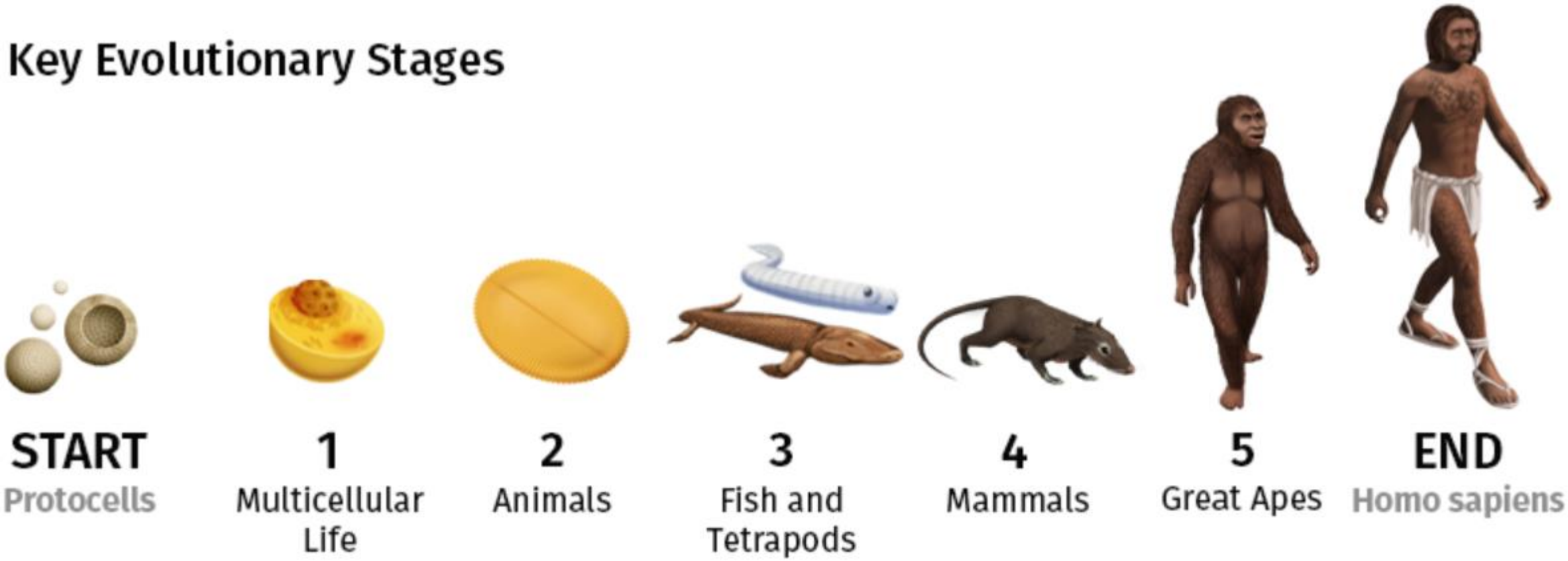
Evolution: Life on earth is one big extended family



https://www.visualcapitalist.com/path-of-human-evolution/



Evolution: Life on earth is one big extended family

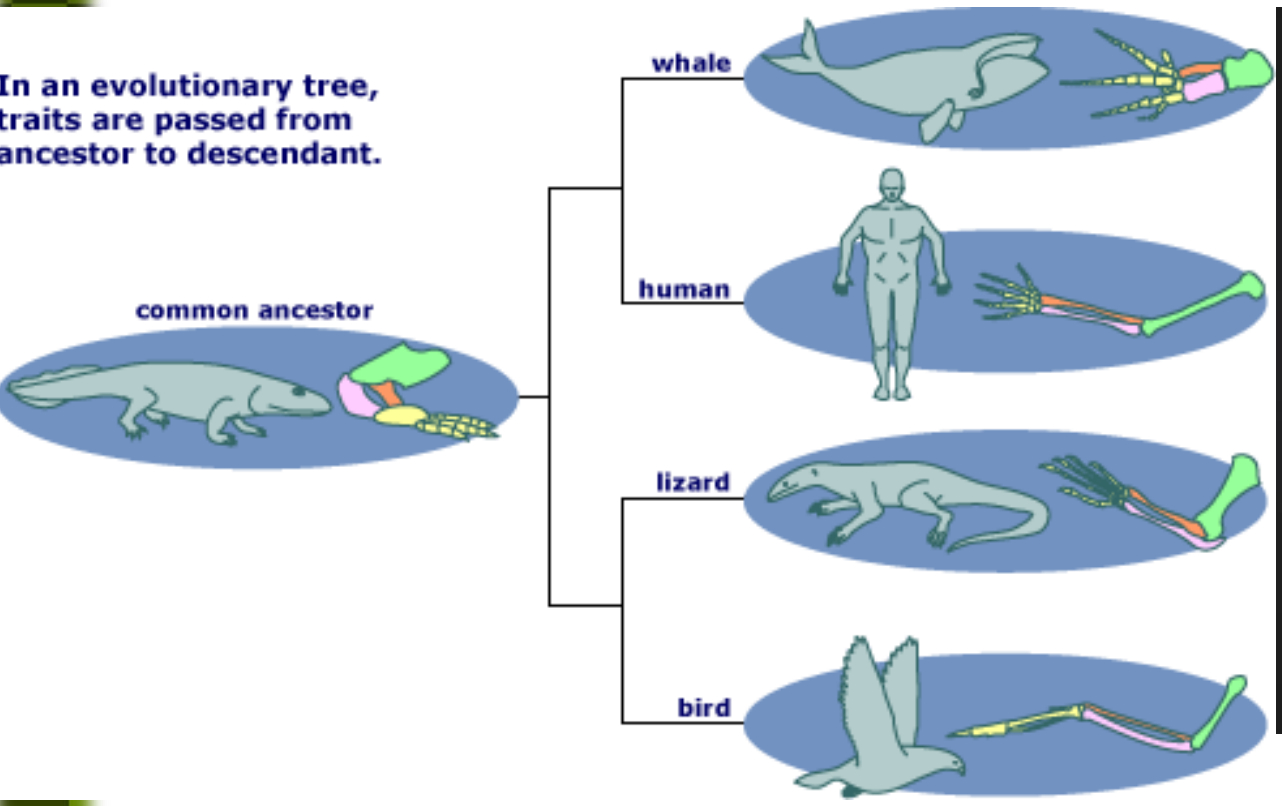


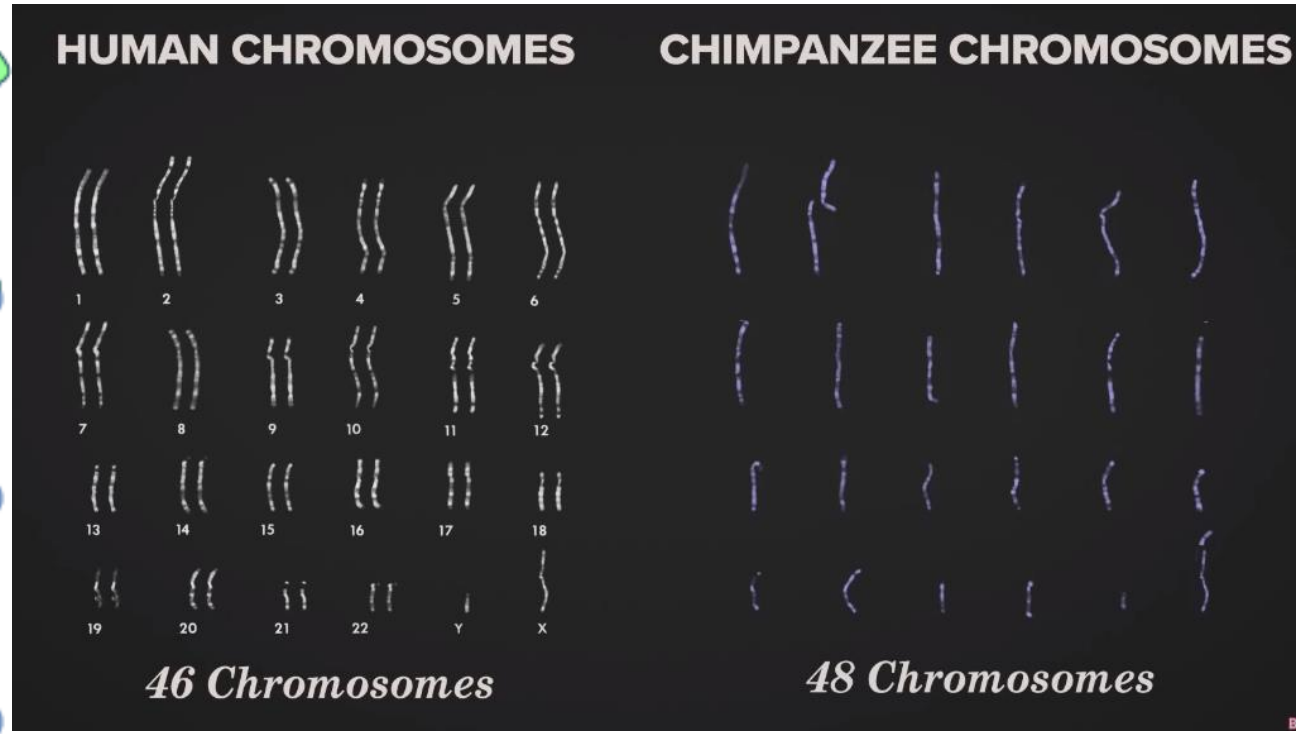
https://www.visualcapitalist.com/path-of-human-evolution/

What is evolution?

- Evolution tries to explain how life must have diversified into multiple forms as we see them today.
- It is change in the inherited traits of a population through successive generations.
    - When living organisms reproduce, they pass on to their progeny a collection of traits.
- Present day organisms are descendants of a common ancestor: Due to multiple heritable modifications.
    - Genetic changes in population of an organism over time.
- It accounts for both unity and the diversity of life
    - Skeletal architecture of limbs
    - DNA

What is evolution?



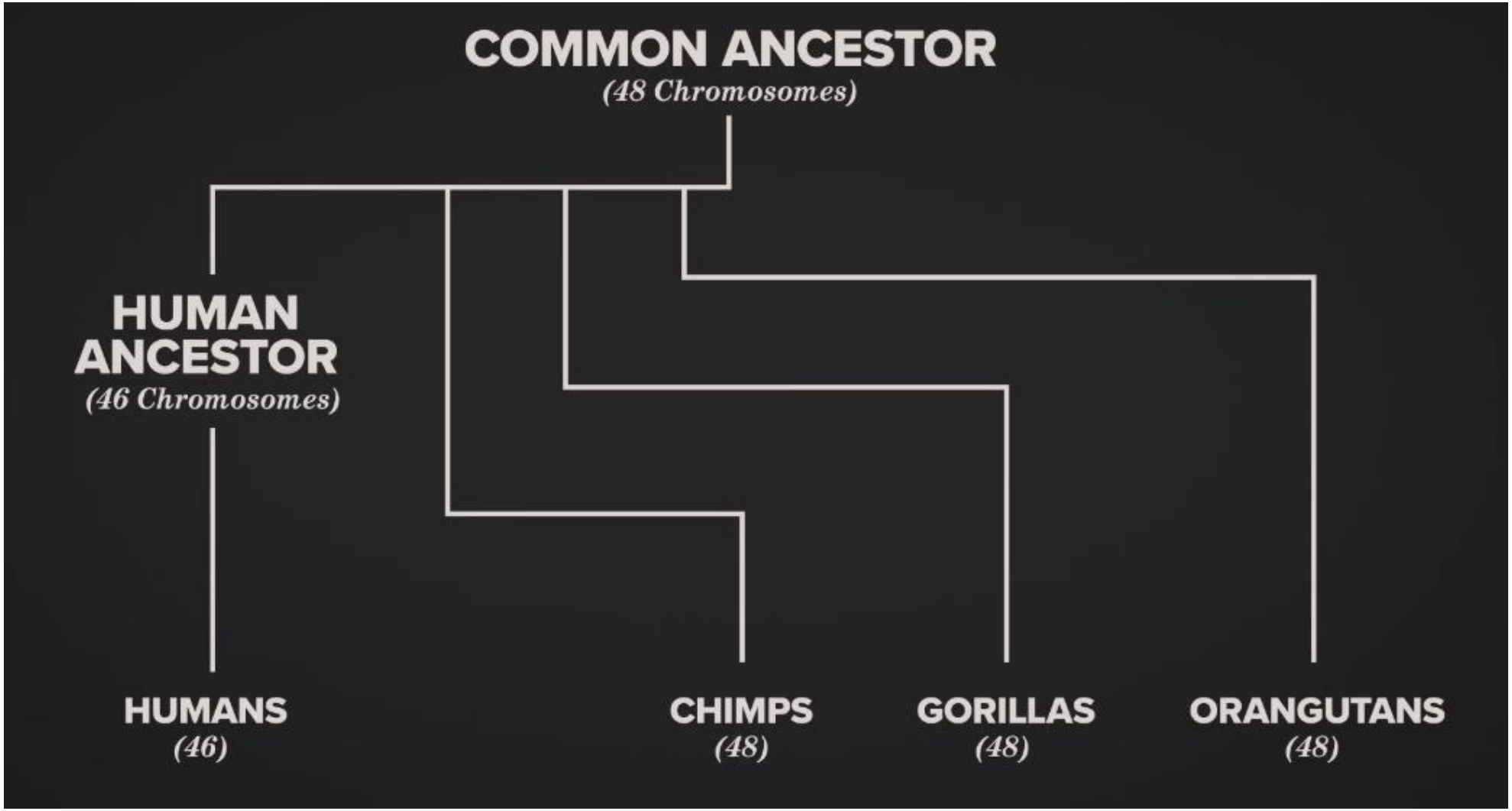


Skeletal architecture of limbs

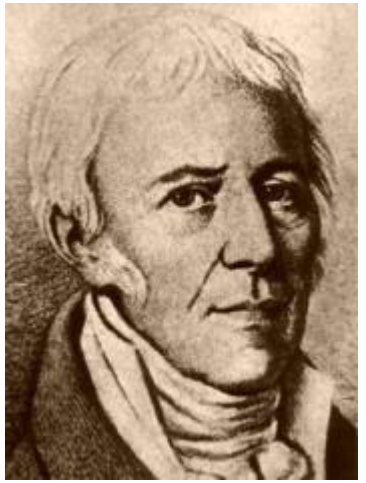
Genetic similarity

https://evolution.berkeley.edu/teach-resources/homologous-tetrapod-limbs-5-of-6/

What is evolution?



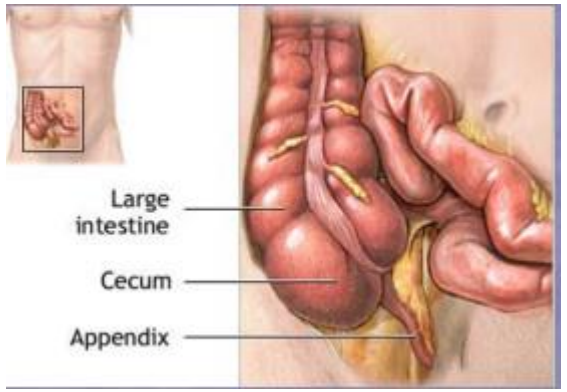
Lamarckian Evolution



Jean-Baptiste de Lamarck
(1744–1829)

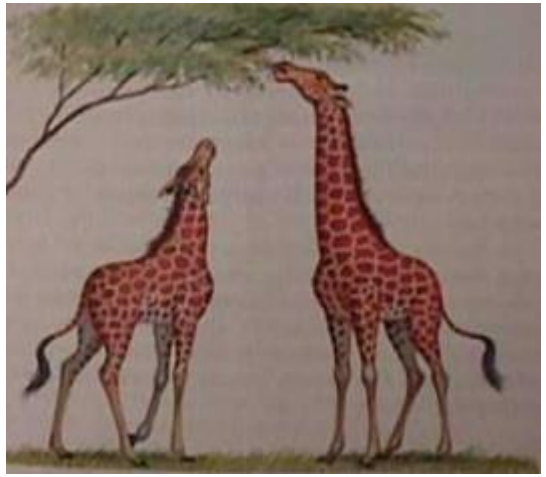
1809 – Lamarck proposed his theory of evolution. Theory is based on 2 hypotheses:

1. Organisms develop traits by the use and disuse of body parts.



Vestigial organs

2. Acquired characteristics – passed from parents to offspring: Inheritance of acquired characteristics.

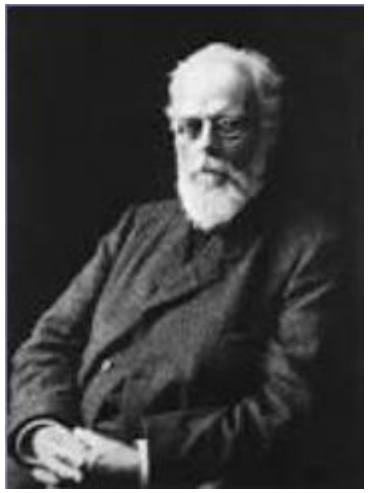


Giraffes had short necks originally and ate grass. Grass died due to climate change, ate leaves of trees As giraffes stretched necks, necks grew. Giraffe's acquired long necks – passed to future generations

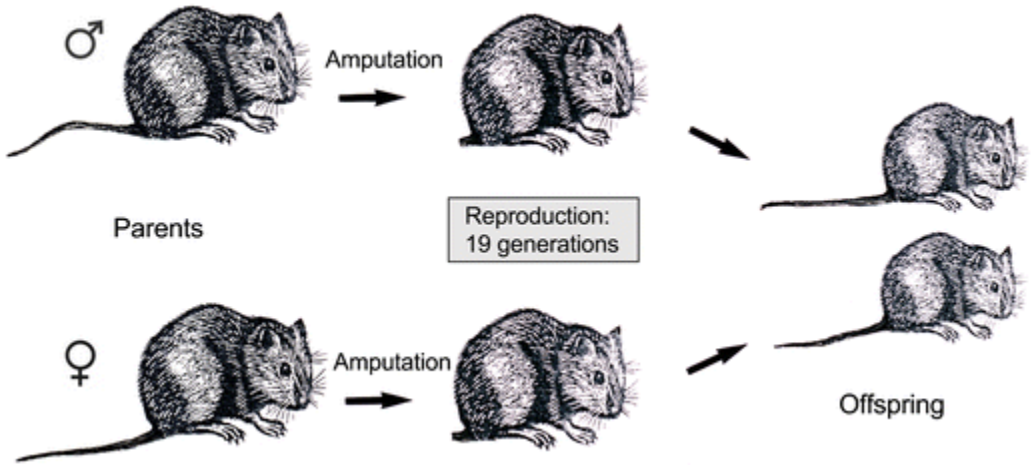
Disproving Lamarck

Characteristics acquired during the lifetime of a parent are not passed onto the offspring:

- In an attempt to disprove Lamarckism, he cut off the tails of mice for several generations.
- The babies born from this line of tailless mice still grew tails as long as their ancestors.
- They still found their tails useful



August Weismann (1834-1913)



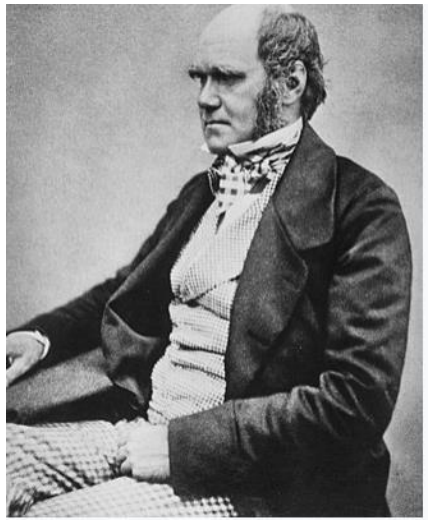
This was not exactly a fair test as the mice had not stopped using their tails in an attempt to adapt to their environment

https://link.springer.com/article/10.1007/s00114-014-1164-4

Voyage of HMS Beagle



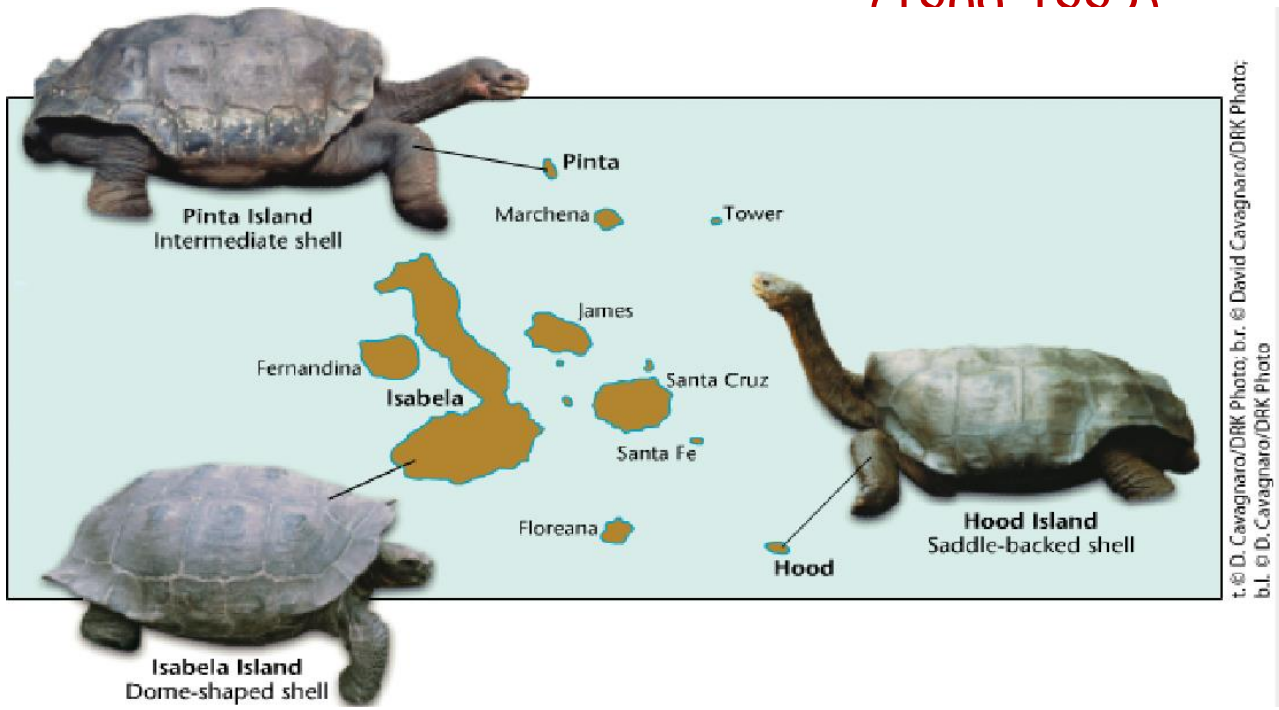
Darwin's Evolution



Charles Darwin
(1809-1882)

Variations The Galapagos Island

- Darwin was fascinated in particular by the land tortoises and marine iguanas in the Galápagos.
- Giant tortoises varied in different ways from one island to another.
- The shape of a tortoise's shell could be used to identify which island a particular tortoise inhabited.



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Darwin's Questions

- Why are there so many species of living things?
- How do new species arise
- Within a population, how does an organism evolve?
- How do organism adapt to their environment?
- How is one species related to another?

Darwin's Observations

Inheritance of traits:

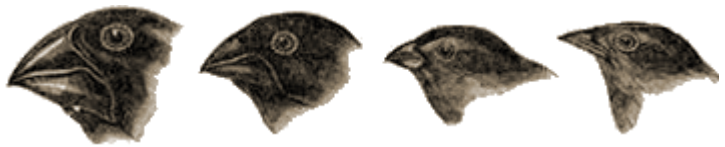
- Variations (differences) of traits occur among members of the same species.
- Examples are different breeds of dogs, horses, cats, etc.
- Differences among the beaks of finches.
- Once these adaptations are selected, these traits are inherited.



https://www.editoriallapaz.org/darwin-en-el-estrado-02-seleccion-natural-hipotesis.html

Darwin's Finches

- These are some of the finches that Darwin studied on his voyage
- Though they are all finches, their beaks distinguish them from one another
- Some finches developed short, strong beaks that made it possible for them to crack nuts
- Some developed long, fine beaks to reach insect larvae in tiny holes
- This is an example of how these finches adapted to their environment to survive



http://darwin-online.org.uk



Darwin's Observations

Overproduction:

- Organisms produce more organisms than can survive and reproduce.
- This ensures that some offspring will survive to reproduce.
- Overproduction is a driving force in natural selection, as it can lead to adaptation
- Examples are fish, insects, frogs, etc



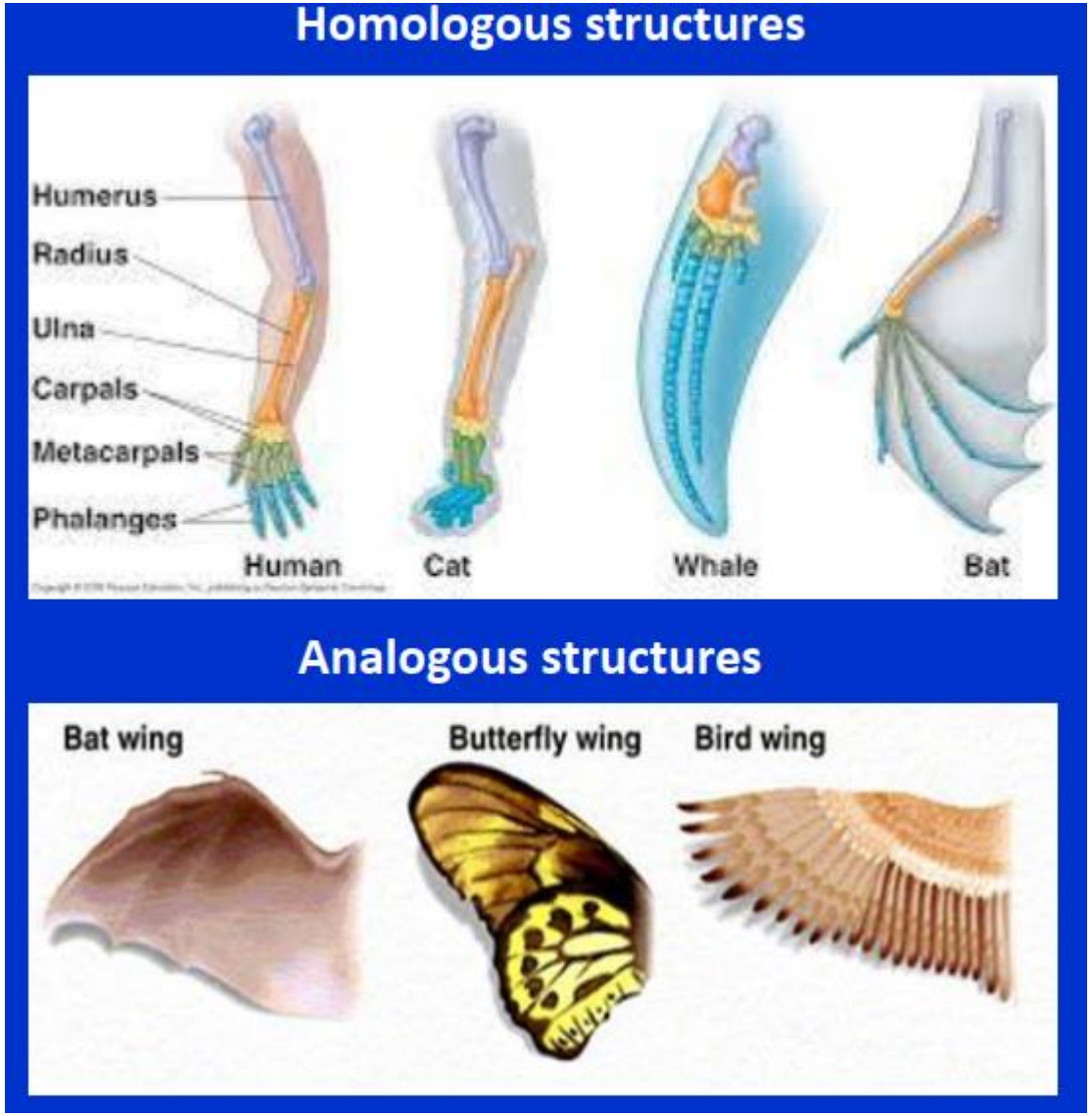
Darwin's Observations

Adaptation and survival of the fittest:

- Adjustments between organisms and the environment are called adaptation.
- Adapted characteristics of organisms that enhance their survival and reproduction in specific environments favours natural selection.

Examples:

- ➢ Homologous structures: Origin same but functions different.
- ➢ Analogous structures: Origin different but functions same.

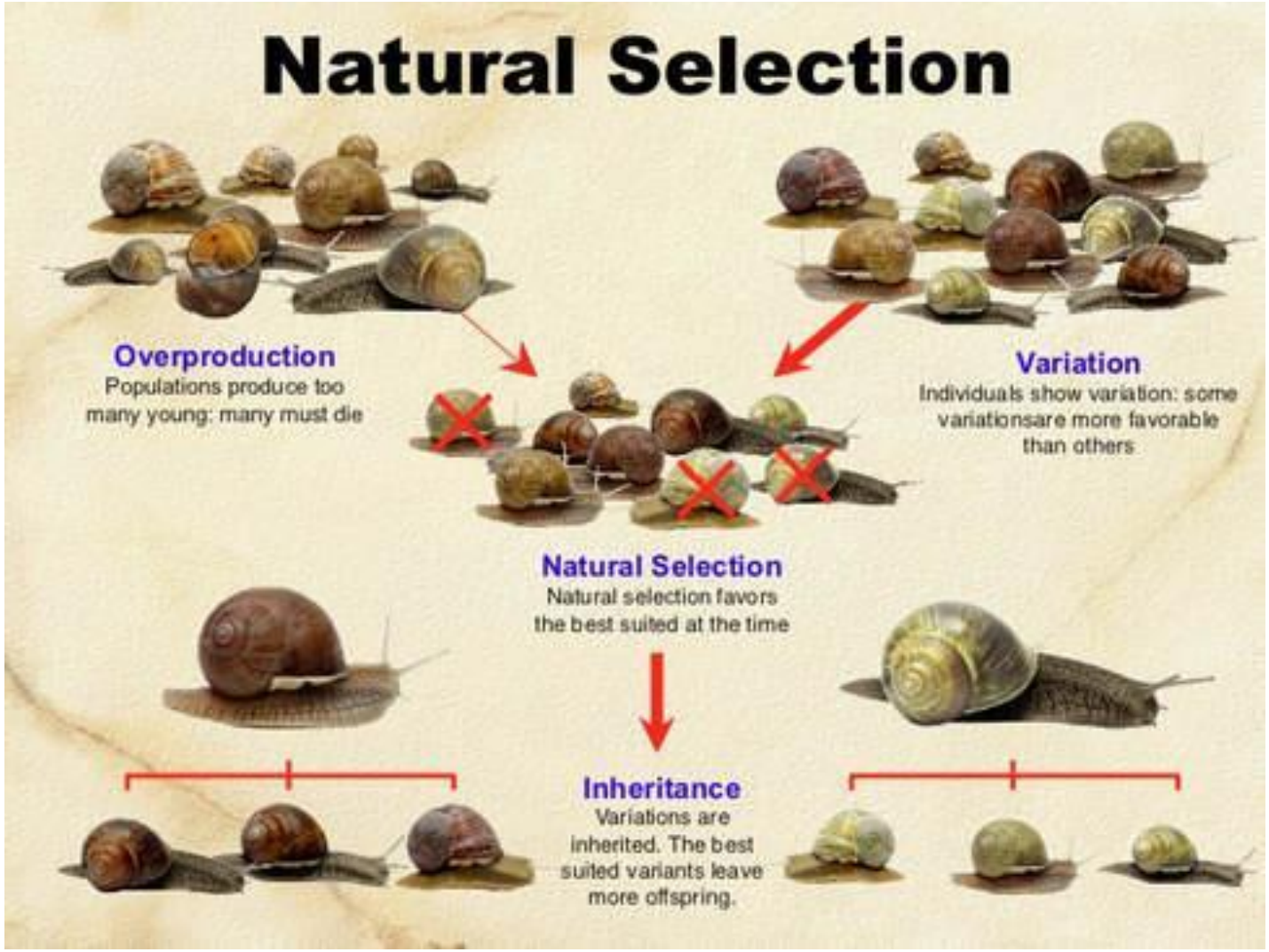


Darwin's Observations: Theory based on 4 main ideas:

The Struggle for Existence:
Members of each species have to compete for food, shelter, other life necessities.

Survival of the Fittest: Some individuals better suited for the environment.

Natural Selection: Environment determines which variations will be selected.



"Individuals with traits which allow them to best adapt to the environment will most likely survive and reproduce"

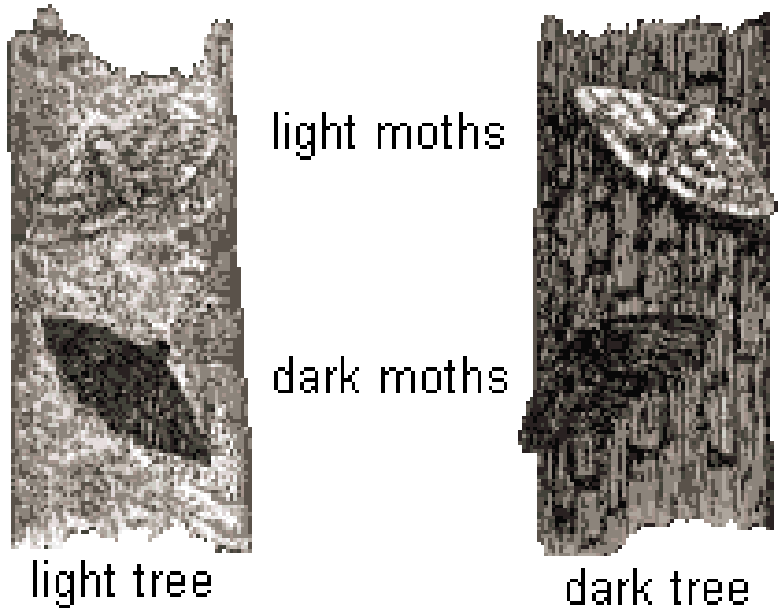
EXAMPLE OF NATURAL SELECTION

Peppered Moth

- Moths can camouflage with trees to avoid being eaten by birds.

- There were light moths and dark moths living near English industrial cities in the 19th century
  - The dark moths stood out on the light colored trees and were more likely to be seen and eaten by birds
    - Thus, there were many more light colored moths than dark colored moths.

- As English factories produced more and more soot, the trees turned a darker color. This was able to camouflage the dark moths, but not the light moths
  - THEN, the light moths stood out on the dark trees and were more likely to be seen and eaten by birds!
    - Thus, there were more dark colored moths than light colored moths.



Because the dark moth was able to camouflage it was able to avoid being eaten.

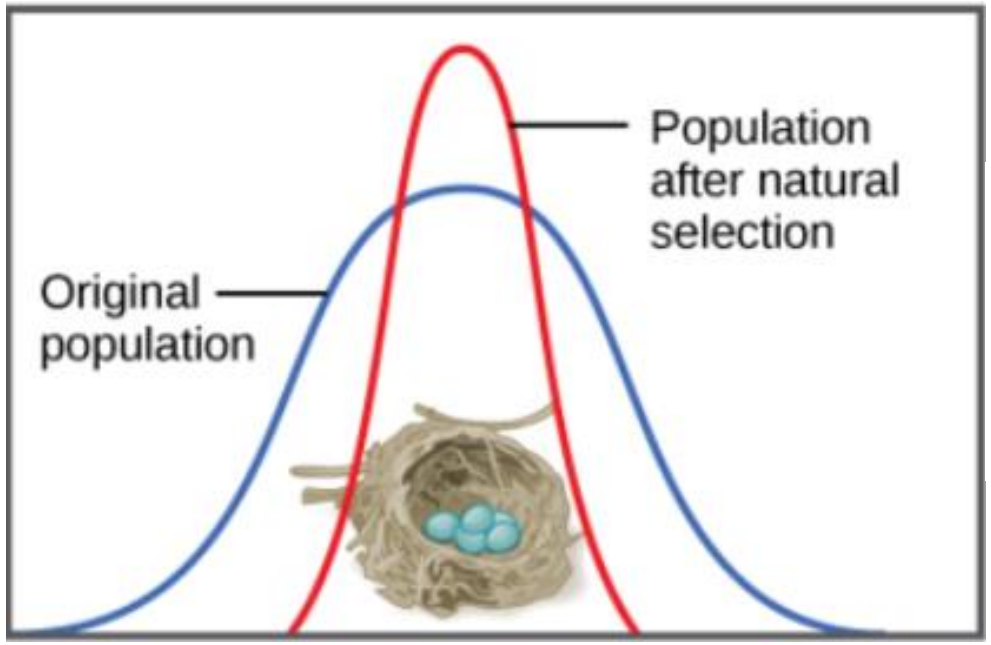
If the light colored moth is not able to adapt to its environment then it will eventually become extinct.

Three types of Natural selection

— Stabilizing selection

— Directional selection

— Diversifying selection



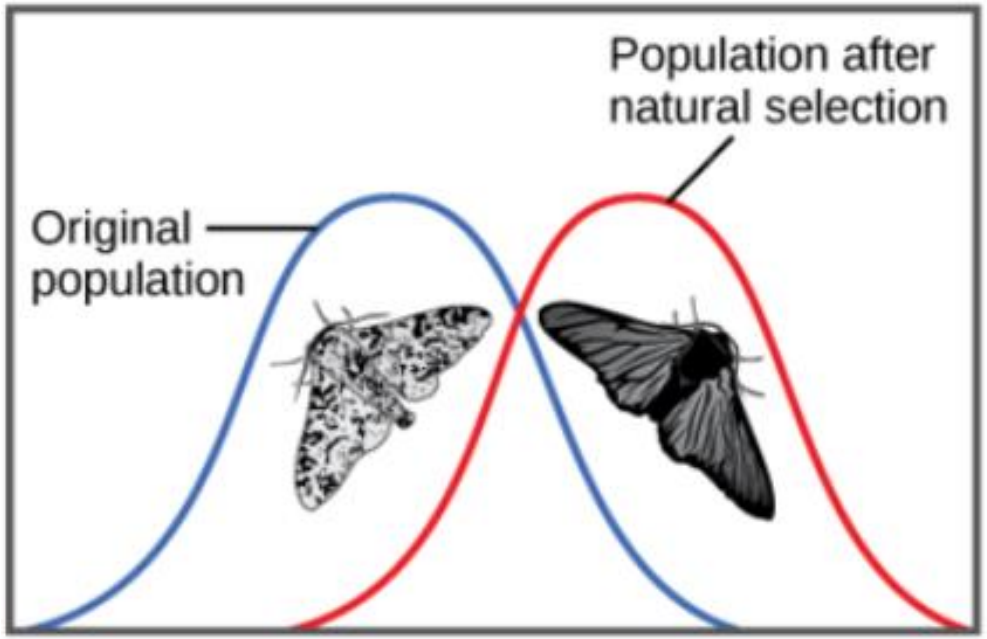
Robins typically lay four eggs, an example of stabilizing selection. Larger clutches may result in malnourished chicks, while smaller clutches may result in no viable offspring.

https://bio.libretexts.org/Courses/Lumen_Learning/Fundamentals_of_Biology_I_(Lumen)/05%3A_Module_2-_History_of_Life/5.16%3A_Adaptive_Evolution

Three types of Natural selection

— Stabilizing selection

— Directional selection



Light-colored peppered moths are better camouflaged against a pristine environment; likewise, dark-colored peppered moths are better camouflaged against a sooty environment. Thus, as the Industrial Revolution progressed in nineteenth-century England, the color of the moth population shifted from light to dark, an example of directional selection.

— Diversifying selection

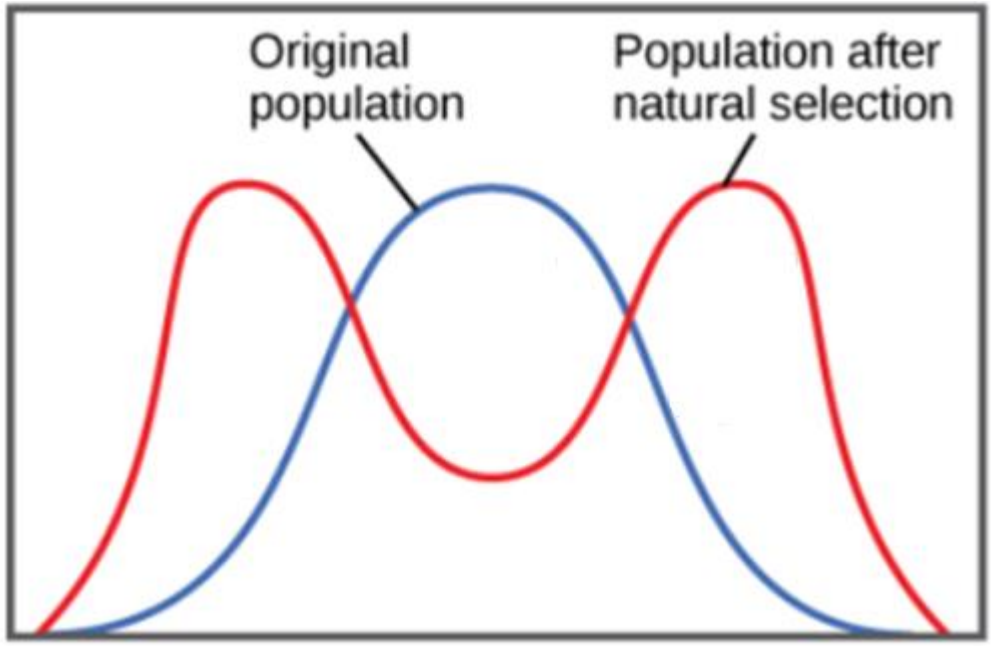
https://bio.libretexts.org/Courses/Lumen_Learning/Fundamentals_of_Biology_I_(Lumen)/05%3A_Module_2-_History_of_Life/5.16%3A_Adaptive_Evolution

Three types of Natural selection

— Stabilizing selection

— Directional selection

— Diversifying selection



An example of disruptive selection is oyster shell color. Light colored oysters blend in with the sand in the shallow water and dark colored oysters blend into the shadows in dark water. Medium colored oysters show up against both backgrounds and thus are not selected for.

DIVERGENT EVOLUTION

Red Fox



Kit Fox



- Divergent evolution is the process of two or more related species becoming more and more dissimilar.
- Example: The red fox and the kit fox
  - The red fox lives in mixed farmlands and forests, where its red color helps it blend in with surrounding trees.
  - The kit fox lives on the plains and in the deserts, where its sandy color helps conceal it from prey and predators. The ears of the kit fox are larger than those of the red fox. The kit fox's large ears are an adaptation to its desert environment
- Similarities in structure indicate that the red fox and the kit fox had a common ancestor. As they adapted to different environments, the appearance of the two species diverged, or became more and more different.

CONVERGENT EVOLUTION

- Convergent evolution is the opposite of divergent evolution
- In convergent evolution, unrelated species become more and more similar in appearance as they adapt to the same kind of environment.
- Frogs and Chameleons are examples of convergent evolution because although they are different amphibians, they have both developed harpoon-like tongues to catch insects





CO-EVOLUTION

- Co-evolution is the joint change of two or more species in close interaction. Predators and their prey sometimes co-evolve; parasites and their hosts often co-evolve; plant-eating animals and the plants upon which they feed also co-evolve.

- One example of co-evolution is pollination of orchids by African moths.

  - These species co-evolve because the moths depend on the flowers for nectar, and the flowers depend on the moths to spread their pollen so that they can reproduce.

- The relationship between the moth and the orchids is mutually beneficial because they help each other survive.





What Darwin's theory fails to answer

- What is the molecular mechanism that causes this variation and how these variations are passed on?

- How do species become extinct?